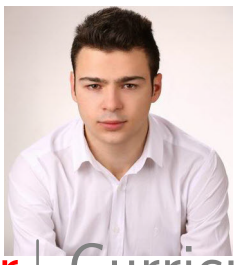

# Erdi Sayar | Curriculum Vitae

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Last Update September 13, 2019

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## WORK EXPERIENCE

### • Student Asistant
RWTH Aachen University
*10/2018 - 04/2019* *Aachen/Germany*

### • Research Asistant
Izmir Institute of Technology
**Teaching**
- ME352 SYSTEM ANALYSIS AND CONTROL
- ME242 APPLIED MATHEMATICS FOR ENGINEERS

*02/2017 - 02/2018* *Izmir/Turkey*

### • Research Asistant
Turkish-German University
*02/2016 - 02/2017* *Istanbul/Turkey*

## Education

### • M.Sc
**RWTH Aachen University**
Electrical Engineering, Information Technology and Computer Engineering
• Systems Engineering and Automation
*2018 - Present* *Aachen/Germany*

### • Ph.D ( Drop-out)
**Izmir Institute of Technology**
Mechanical Engineering
*2016 - 2018* *Izmir/Turkey*

### • B.Sc
**Bochum University of Applied Sciences**
Mechatronics Engineering
*2013 - 2014* *Bochum/Germany*
***GPA :*** 1.36/1.00
***Thesis :*** Three phase asynchronous motor dynamic mathematical model and control Simulink
**Advisors :**Prof.Dr.-Ing Peter Brychta,Prof. Dr.-Ing. Arno Bergmann

### • B.Sc
**Kocaeli University**
Mechatronics Engineering
*2010 - 2015* *Kocaeli/Turkey*
***GPA :*** 3.82/4.00
*Ranked first in the faculty of engineering.Graduated with summa cum laude.*

## Engineering Skills

Matlab & Simulink | Python | C & C++ | ROS | Maple | Autocad | SolidWorks | Ansys | Visual Basic | Proteus | PSPICE | Eagle | LaTeX

## Interests

Surf, Swimming

## Projects

• Using Cosimir Software detect objects colors and sort them with respect to colors [Robotic Lecture] • Using Solidworks software , designing "Mechanical Press"mechanism and displacement, velocity and acceleration analysis [Theory of Machine Lecture ] • Fuzzy Logic Controller and PID Controller Design for Aircraft Pitch Control [Fuzzy Logic Controller Lecture] • Ansys Analysis of Intermittent worm wheel with cam [Finite Elements Lecture]] • Using Fanuc 6 Axes Robot and programming with Karel Programming language for pick and place task [Robotic Lecture] • Simulation of Asynchronous motor in Matlab/Simulink [Bachelor Thesis] • Simulation of Scalar and Vector Control of Asynchronous Motor in Matlab/Simulink [Bachelor Thesis] • Line Following Robot and Obstacle Avoidance Robot by using Arduino [Sensor Techniques Lecture] • Cryptocurrency Trading Robot Software in Python

## Achievements

• Ranked 1st in the Faculty of Engineering ,Kocaeli University (over 1000 students)

• Ranked 1st in Mechatronics Department ,Kocaeli University

• (TUBITAK)Turkish National Science Foundation 2211/A PhD scholarship

• Outstanding Teaching Assistant ,Izmir Institute of Technology

## Publications & Certificates

• Fuzzy Logic Controller and PID Controller Design for Aircraft Pitch Control at IFFTOM Conference
• Assistive Handwriting Haptic Mechanism Using Deep Learning Speech Recognition Accepted by IFToMM committe in Taiwan however, I couldn't publish due to financial problems.
• ROS Summer School Certificate at FH Aachen

## Languages

**Turkish Native**
**English TOEFL 82/120** ● ● ● ● ○
**German Telc Deutsch Zertifikat** ● ● ● ● ○

## Area of Interest

•Robotic ,Control, Computer Vision, Machine Learning

## Taking Lectures relating to Area of Interest

- Machine Learning $[R,E,M]$
- Artificial Intelligence $[R,E,M]$
- Machine Learning Laboratory $[R,E,M]$
- Advanced Robotic Kinematics And Dynamics $[R,E,M]$
- Multi Body Dynamic $[R,G,M]$
- Artificial Neural Network $[R,G,M]$
- Robotic Man-Machine Interaction 2 $[R,G,M]$
- Modern Control Systems $[R,E,M]$
- Rapid Control Prototyping $[R,G,M]$
- Multi Media Communication 2 $[R,E,M]$
- Principles of Robotics $[I,E,P]$
- Advanced Robotic Kinematics $[I,E,P]$
- Digitial Image Processing $[I,E,P]$
- Introduction to Fuzzy Logic $[K,E,B]$
- Robotics and Automation Systems $[K,T,B]$
- Fundamentals of Image Processing $[K,E,B]$
- Automatic Control $[K,T,B]$

$[X,Y,Z] \quad X \in \{R,I,K\} \quad Y \in \{E,G,T\} \quad Z \in \{P,M,B\}$ *where X is name of university, Y is the instruction language, Z is the type of education.*
$R = RWTH\ Aachen\ University, I = Izmir\ Institute\ of\ Technology,\ K = Kocaeli\ University$
$E = English,\ G = German,\ T = Turkish$
$P = Ph.D,\ M = Master,\ B = Bachelor$

## Taking Lectures relating to Area of Interest

### Machine Learning

Bayes Decision Theory,Probability Density Estimation Maximum Likelihood Approach,Kernel Methods,K-Means Clustering,Linear Discriminant Function,Logistic Discriminant Function, Least-Square Classification,Softmax Regression,Support Vector Machine,Adaboost,Neural Networks,Optimization Convolutional Neural Network,Recurrent Neural Networks.

### Artificial Intelligence

Problem-Solving,Search and Games,Knowledge, Reasoning and Planning, Learning and Neural Network.

### Machine Learning Laboratory

Supervised Learning,Feature Extraction,Unbiased Evaluation,Neural Networks, Backpropagation,Deep Learning,Segmentation and Optimization.

### Advanced Robotic Kinematics And Dynamics

Orientation and Homogenous Transformation using Rotation Matrices ,Axis-Angle Representation,Quartenions,Direct and Inverse Kinematics, Forward and Inverse Differential Kinematics,Robot Dynamics with Lagrange and Newton-Euler, Identification Approach for Dynamic Parameters,Trajectory and Path Planning.

### Multi Body Dynamic

Modeling and Kinematic, Equations of Motion and Linearization, Equation of States and Harmonic excitation.

### Artificial Neural Network

Supervised and Unsupervised Learning,SVM , Self-organizing Maps and Associative Memory,Dynamics of Pulsneuron.

### Robotic Man-Machine Interaction 2

Dynamic models of industrial Robots ,Machine Learning,Represantation and Processing of Knowledge,Fuzzy Logic,Newton-Euler and Lagrange.

### Modern Control Systems

State Control, Optimal Control, Kalman Filter, Nonlinear Control, Lyapunov Theory.

### Rapid Control Prototyping

Physical Modelling,Identification of Dynamic Systems,Discrete and hybrid simulation with Stateflow.

### Multi Media Communication 2

Feature Extraction(Edge,Texture,Motion), Feature Transform and Classification.

### Principles of Robotics

Spatial Forward and Inverse Kinematics,Velocity and Acceleration Analysis.

### Advanced Robotic Kinematics

Kinematic and Dynamic Control of Serial Robotis.

### Digitial Image Processing

Image Enhancement, Segmentation,Wavelet Transform, Color Image Processing,Morphology, ImageCompression, Camera Models.

### Introduction to Fuzzy Logic

Fuzzification and Defuzzification methods,Fuzzy Logic Controller.

### Robotics and Automation Systems

Resolving Langrange-Euler and Newton-Euler equations for robots,Developing the trajectory of robots.,Calculating Jacobian matrix.

### Fundamentals of Image Processing

Spatial and Frequency domain filtering,Basic morphological operations,Basic segmentation methods.

### Automatic Control

Representing the control systems based on their transfer functions and state space,Explaining the controllability and observability concepts,Recognizing the response of control systems at time and frequency domain,operation principles of PID algorithms and their industrial applications,Realizing the simulation of control systems at Matlab/Simulink environment.